

gdTAI V4.6 NK-exclusion evaluation: how it rules out NK, and how confident we can be

Date: 2026-08-20 · **Status:** independent read-only evaluation of the frozen V4.6 development model using the frozen whole-atlas per-cell cache (5,933,312 cells, 40 sources). No H5AD, model, threshold, or registry entry was modified.

Question. The whole-atlas report shows V4.6 predictions with ~0% “strict likely-NK” (232 of 259,154 calls), versus 24.1% for V2 high-F1. Is that real, and how confident is it?

Short answer. The NK exclusion is real and strong — but “0%” is only literally true on the *strictest* NK definitions. On fully independent (never-developed) sources: strict expression-rule NK call rate is **0.004%**, author-annotated NK **0.53%**, NK-union **0.44%** — 60–70× lower than V2 on identical cells. On the hardest biologically ambiguous stratum (shared cytotoxic program), V4.6 still calls **11.9%** — not zero, and that is where the honest uncertainty lives. The NK-annotated cells V4.6 *does* call are enriched for $\gamma\delta$ -TCR evidence and TRDC expression, i.e., they are plausibly true $\gamma\delta$ T mislabeled as NK rather than model errors. Confidence: high for definitionally-clean NK, bounded and unresolvable for the cytotoxic-ambiguous zone without new gdTCR data.

1. How V4.6 rules out NK — the mechanism

V4.6 has **no NK-marker features and no hard NK veto** (`hard_single_nk_gene_exclusion: false`). Stage-2 features (182) are purely **receptor architecture**: 145 individual TCR/CD3/CD4/FOXP3 genes plus 37 engineered receptor aggregates (TRDV counts/max, TRDC×TRDV joint, δ -minus- $\alpha\beta$ receptor sums, conditional- γ terms). NK exclusion emerges from three structural facts:

1. **Receptor-grammar features:** most NK cells have no coherent TRDV/TRDC receptor signal — they simply have nothing for the classifier to fire on. The 0.09% atlas figure is mostly this: absence of the *positive* pattern, not detection of the *negative* one.
2. **Stage-1 T-lineage gate** (36 T-lineage genes, very low threshold 0.0095 — a high-recall gate): NK cells lacking a CD3/T-lineage program are filtered before Stage 2.
3. **Tiered NK negatives in training** (`nk_tier1` 28,692 + `nk_tier2` 80,541 across 30 development sources, including GSE169246 author-NK): the model was explicitly taught the NK boundary — which is also why the development-cohort NK audits are partly circular.

2. Is the 0% circular? — development-exposure split

NK stratum	never-developed (independent)	development-negative (circular)	development-other (training-adjacent)	V2 high-F1 (all)
strict expression NK (n=313k)	0.004% (84k cells)	0.031%	0.044%	28.8%

NK stratum	never-developed (independent)	development-negative (circular)	development-other (training-adjacent)	V2 high-F1 (all)
author-annotated NK (n=168k)	0.53% (106k cells)	~0%	8.4%	30.0%
NK union (n=406k)	0.44% (126k cells)	0.03%	2.2%	28.6%
TRDC+TRDV- (n=359k)	1.14% (80k cells)	0.84%	2.0%	51.1%
shared cytotoxic ambiguous (n=980k)	11.9% (160k cells)	7.7%	11.5%	19.2%

Read-out:

- The near-zero rates in the 8 development-negative extension cohorts are **circular by construction** (the model was trained to avoid exactly those NK cells); they are a consistency check, not evidence.
- The **never-developed** column is the honest measurement: 9 atlas sources V4.6 never saw in any role. Strict-expression NK 0.004%, author-NK 0.53% (driven by GSE206325 at 0.52%; most sources 0.00%), NK-union 0.44%. The untouched-BALF result (0/254 author-NK; 2/1,078 NK-union in the atlas cache) agrees.
- The whole-atlas headline “232 strict NK (0.09%)” is therefore *directionally* honest but defined narrowly; the defensible claim is **≤0.5% on clean NK definitions**, not zero.

3. Score margins — how fragile is the exclusion?

NK-union (never-developed)	value
score p50 / p90 / p99	0.00 / 0.46 / 0.78
cells scoring ≥ 0.8 (near-miss vs threshold 0.894)	0.76%
cells ≥ threshold (called)	0.44%

In independent sources, 99% of NK-union cells score ≤0.78 — a ~0.11 margin below the operating threshold, and only 0.76% sit in the near-miss band. This is a *confident* exclusion, not a knife-edge one; it is far healthier than the V4.3 pattern (positive median below threshold). The tail exists (a handful of NK-annotated cells score ≈1.0), and those are analyzed next.

4. The cells V4.6 calls among NK-annotated populations look like real γδT

- Of the 200 highest-scoring NK-union cells atlas-wide: **152 have γδ-TCR evidence, 186 are TRDC+**, and V2 calls 190 of them too. These are not random NK leakage; they are exactly the cells the label system cannot adjudicate (NK-annotated but TCR/TRDC-positive — consistent with the kimi round-3 finding that author-NK labels in non-gdTCR-assayed sublibraries are contaminated with true γδT).

- On cells that are *both* author-NK and $\gamma\delta$ -TCR-positive (n=4,983), V4.6 calls 57.0% — *more* than V2 (52.4%). When TCR evidence exists, V4.6 overrides the NK annotation rather than obeying it. That is the correct behavior given what we know about NK label noise.

5. The honest ceiling: the shared-cytotoxic stratum

On cells expressing the NK/ $\gamma\delta$ T-shared cytotoxic program without other adjudicating evidence (980k cells), V4.6 calls 11.9% (never-developed) — better than V2 (19.2%) but far from zero. This stratum mixes true $\gamma\delta$ T (cytotoxic by nature), true NK, and NK-like $\gamma\delta$ T; **no RNA-only model can resolve it, and its call rate should be read as “uncertainty budget,” not error**. This is the stratum the abstain-zone policy exists for: V4.6’s contract already defines an `uncertain_gdt` band between the two thresholds.

6. Stage-1 vs Stage-2 division of labor (added 2026-08-20)

Stage-1 rejections are encoded in the cache as a sentinel score of -1 , so the two stages’ contributions can be measured exactly:

stratum	n	Stage-1 rejects	Stage-2 calls among passers	net called
author-NK	168,483	62.2%	9.1%	3.4%
strict expression NK	313,360	68.8%	0.10%	0.03%
NK union	406,189	64.1%	4.1%	1.5%
TRDC ⁺ TRDV ⁻	358,695	58.9%	4.1%	1.7%
shared cytotoxic ambiguous	979,723	19.4%	13.7%	11.1%
$\gamma\delta$ -TCR ⁺ (positives)	68,811	3.3%	73.9% called	71.4%
paired- $\alpha\beta$ -TCR ⁺	1,938,158	23.8%	0.39%	0.30%

So the user’s inference is correct: **Stage 1 does the majority of NK removal (~two-thirds of NK-union cells), while Stage 2 removes most of what remains** — and it does so while rejecting only 3.3% of $\gamma\delta$ -TCR⁺ cells (the gate’s high-recall design works). Note that the Stage-1 gate is one-sided: it asks “does this look like a T cell” from 36 T-lineage genes and contains no NK markers, so NK cells fail it by *absence* of the T program; TRDC⁺ NK cells can pass it (TRDC/TRGC1/TRGC2 are gate genes) and are then handled by Stage 2, which is exactly the stratum where Stage 2 earns its keep (TRDC⁺TRDV⁻: 58.9% gated, 4.1% of passers called).

Stage-1 gene panel (36): core TCR complex — CD3D, CD3E, CD3G, CD247, TRAC, TRBC1, TRBC2, TRDC, TRGC1, TRGC2, TRAT1; T-lineage identity — BCL11B, LCK, LAT, CD2, CD5, CD6, TCF7, LEF1, THEMIS; co-receptors — CD4, CD8A, CD8B; naive/helper program — IL7R, CCR7, SELL, LTB, MAL; Treg program — FOXP3, IL2RA, CTLA4, TIGIT, IKZF2; tissue/effector state — KLRB1, ZNF683, RUNX3.

7. Verdict

1. **The NK exclusion is real:** ~0.4–0.5% call rate on independent author/expression NK with comfortable score margins, 0/254 on the one untouched cohort, 60–70× better than V2, and the called NK-annotated cells are TCR/TRDC-enriched (likely mislabeled $\gamma\delta$ T, not leakage).
2. **“0%” is marketing of a narrow stratum:** quote it as “ $\leq 0.5\%$ on clean NK definitions, 0.004% on strict expression NK” — and never pool the development-negative cohorts into the claim (circular).
3. **The unresolvable remainder is the cytotoxic-ambiguous zone (~12% called):** bound it, abstain on it, and do not iterate further against it without new gdTCR data.
4. Confidence grade: **high** for clean-NK exclusion, **moderate** for TRDC⁺TRDV⁻ (1.1%), **inherently bounded** for cytotoxic-ambiguous cells.

Appendix: artifacts

- Script: `workflows/gdtai/run_v4_6_nk_evaluation.py`
- Tables: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_6_nk_evaluation/`
(`nk_strata_evaluation.csv`, `nk_union_by_source.csv`, `nk_union_top_scoring_cells.csv`)
- Inputs (read-only): frozen whole-atlas prediction cache, V4.6 development partition manifest, V4.6 model contract.